

Towards Noise-Aware Quantum Compilation for Hamiltonian Simulation

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Introduction

Goal: Improve quantum compilation of fermionic Hamiltonians, a class of problems modeling molecular and material dynamics, for NISQ and early fault-tolerant devices.

Challenge: These devices have varying coupling qualities. Noise-unaware compilation could result in suboptimal circuits.

Contribution: We accelerate and introduce noise-aware coupling selection to Treespilation, a state-of-the-art compilation framework for fermionic Hamiltonians.

Prior Work: Treespilation

Fermionic Hamiltonians can be expressed as

$$H_f = \sum_{ij} c_{ij} m_i m_j + \sum_{ijkl} c_{ijkl} m_i m_j m_k m_l,$$

where m_i 's are the Majorana operators.

To simulate the system, a fermion-to-qubit (F2Q) mappings is used to map each Majorana operator to a Pauli string:

$$m_i \mapsto S_i \in \pm\{I, X, Y, Z\}^{\otimes n}.$$

Many F2Q mappings can be drawn as a ternary tree (TT):

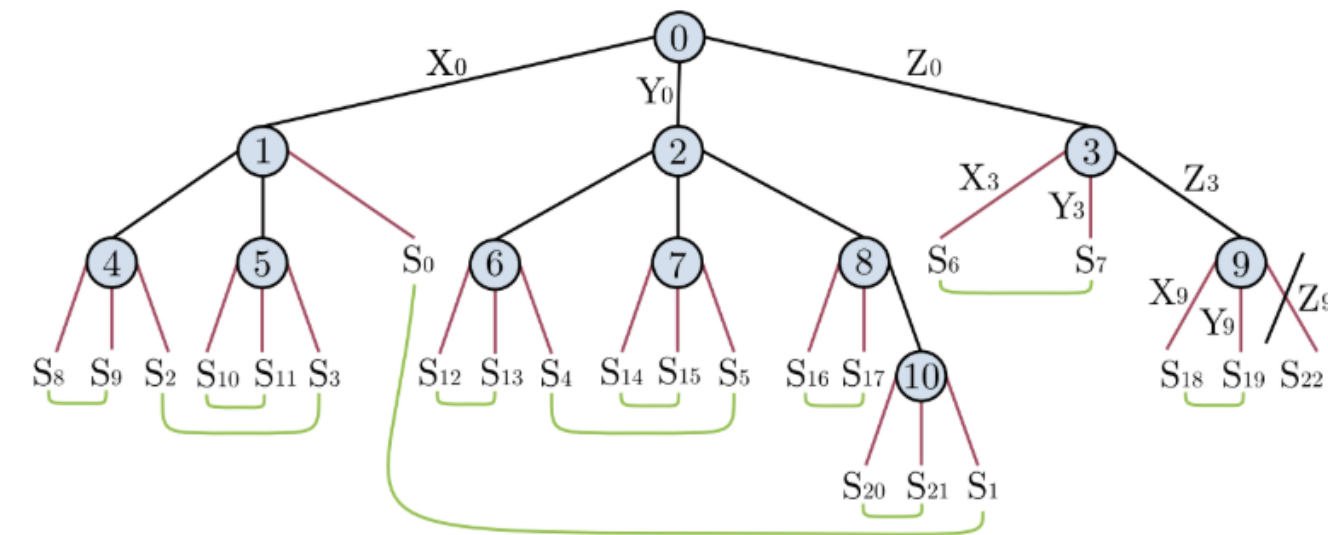


Figure: A ternary tree for F2Q mapping. Each S_i is obtained by traversing from the root to the label, collecting the Pauli letters. Image from [1].

Example: mapping product $m_4 m_5$:

$$S_4 = Y_0 X_2 Z_6, S_5 = Y_0 Y_2 Z_5 \Rightarrow S_4 S_5 = i Z_2 Z_5 Z_6.$$

If the TT is a subgraph of the coupling graph, qubit routing becomes unnecessary. The tree shape directly corresponds to gate placements!

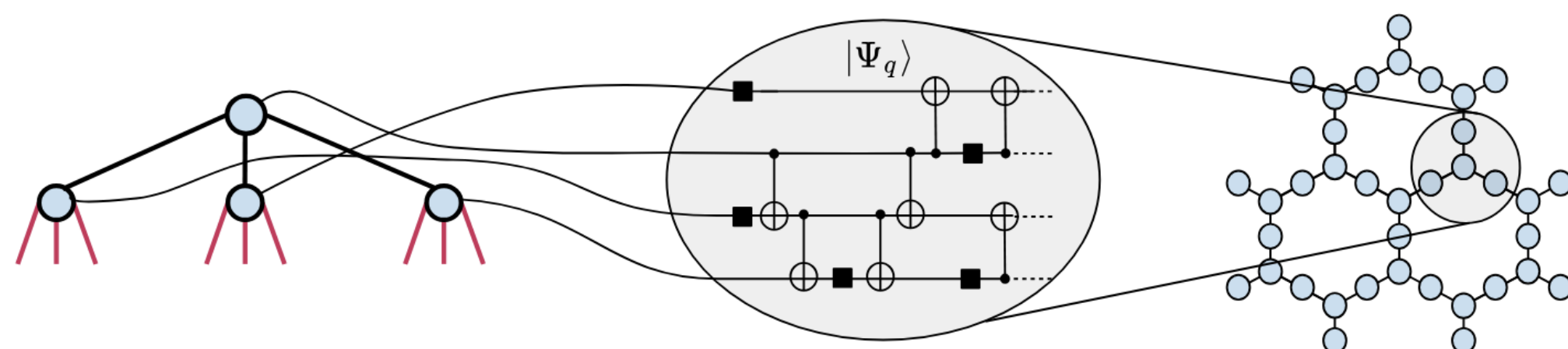


Figure: Hardware-compliant TTs direct gate placements. Image from [1].

Accelerating Ternary Tree Optimization

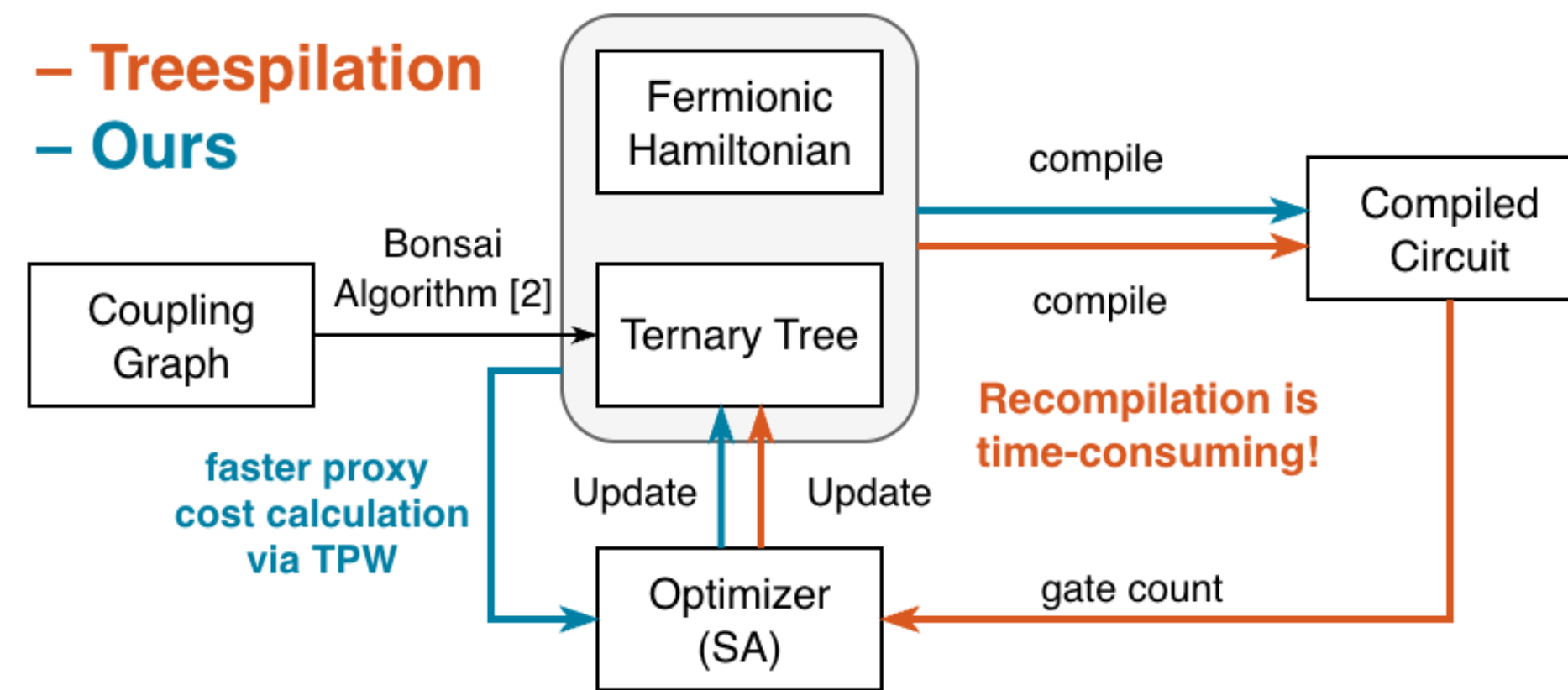


Figure: Accelerating Treespilation with proxy metric.

Treespilation uses simulated annealing with gate count as the cost function, which requires repeated recompilation. We replace this with Total Pauli Weight (TPW) as a proxy:

$$\text{TPW}(H_q) = \sum_{P \in H_q} \sum_{i=1}^n \mathbb{1}(P_i \neq I).$$

TPW is cheap to compute and does not require compilation.

Noise-Aware Coupling Selection

Treespilation places most CNOT gates on tree-edge couplings. However, it is not aware of coupling gate fidelity.

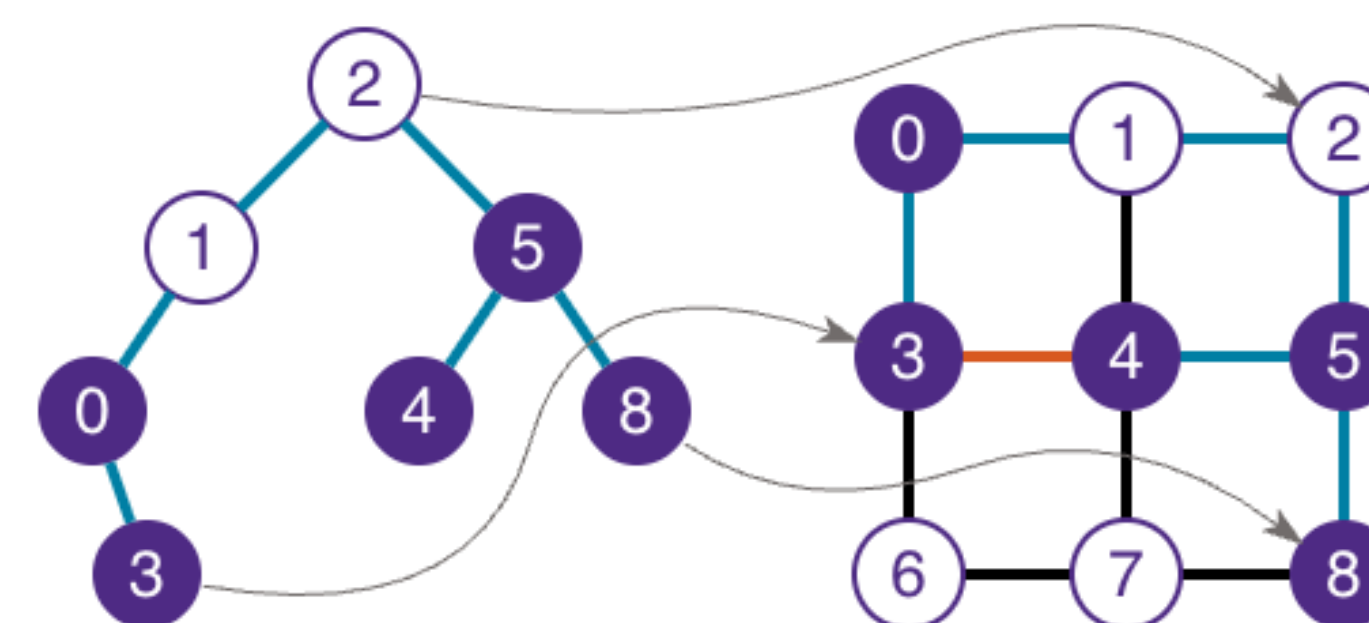


Figure: Synthesizing Pauli Rotation Terms in presence of unreliable coupling. Red: Bad coupling. Cyan: couplings used as tree edges.

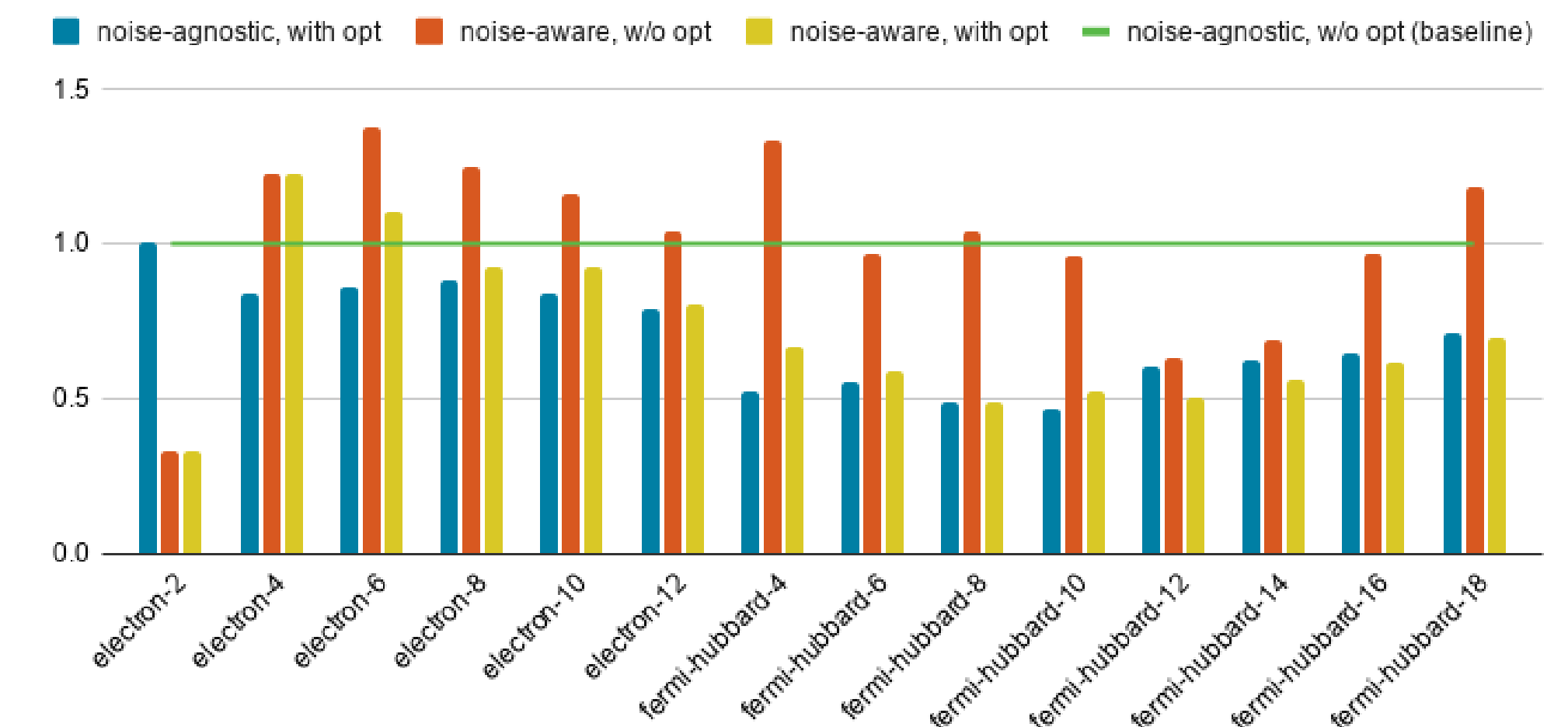
Using a log-success-rate $\text{cost}(i, j) = -\log_2(1 - \epsilon_{ij})$, we enable noise-awareness by

- Weighted Bonsai algorithm [2] and tree perturbation rules
- Finding optimal MST for CNOT circuit synthesis

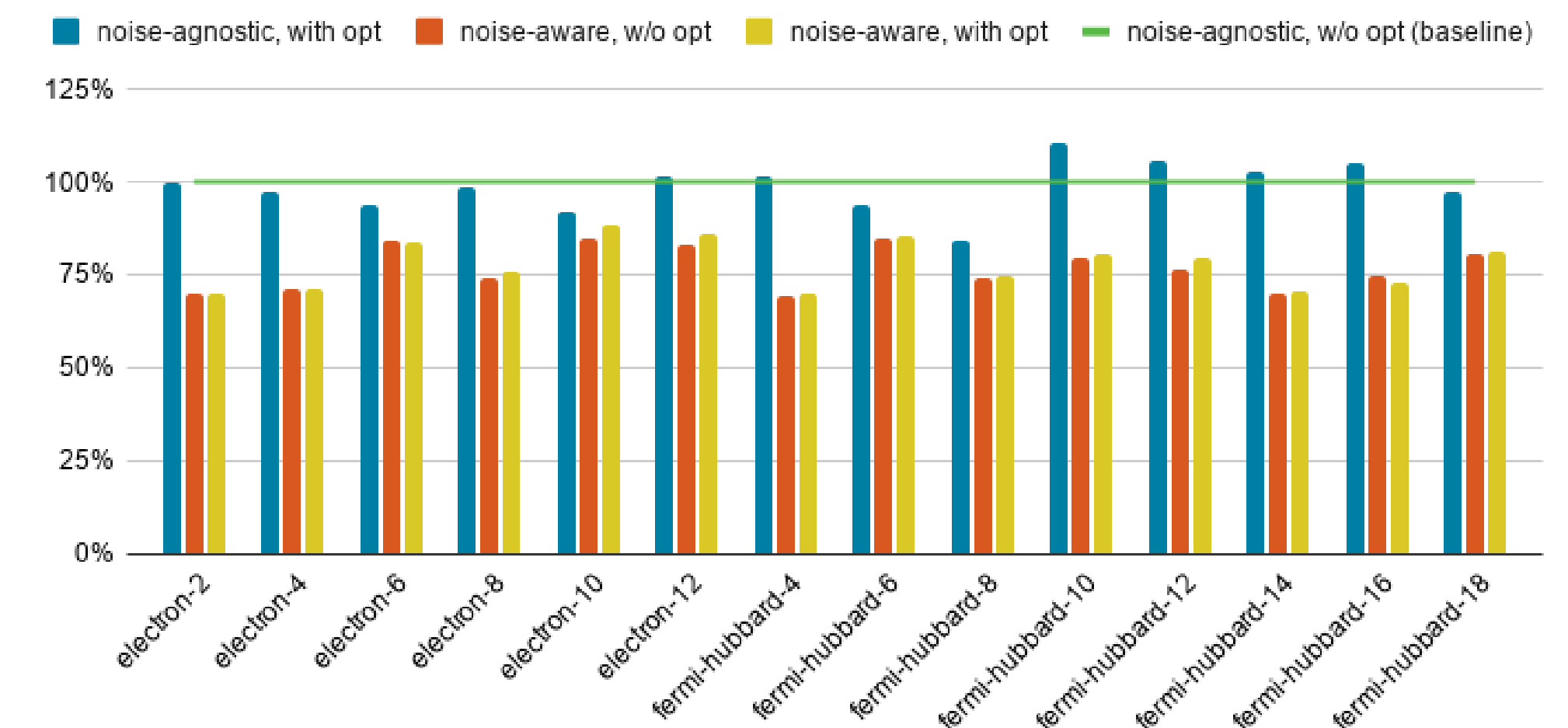
[1] A. Miller, A. Glos, and Z. Zimborás, "Treespilation: architecture- and state-optimised fermion-to-qubit mappings," npj Quantum Inf, vol. 12, no. 1, p. 26, Jan. 2026, doi: 10.1038/s41534-025-01170-2.

Results

On average, the proxy TPW metric reduces the circuit gate count by **28%**. Noise-aware coupling selection reduces geomean coupling gate error rate in circuits by **25%**.



(a) Relative gate counts



(b) Rel. geomean coupling gate error rates

Figure: Relative gate counts and error rates for different Treespilation configurations. Baseline: noise-unaware Treespilation w/o tree optimizations.

Future Work

- Simulating compiled circuits to test end-to-end fidelity
- Developing proxy costs that are aware of other sources of noises, e.g., qubit coherence times
- Mapping heavily used tree edges to good couplings
- Optimizing for circuit depths

[2] A. Miller, Z. Zimborás, S. Knecht, S. Maniscalco, and G. García-Pérez, "Bonsai Algorithm: Grow Your Own Fermion-to-Qubit Mappings," PRX Quantum, vol. 4, no. 3, p. 030314, Aug. 2023, doi: 10.1103/PRXQuantum.4.030314.